

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 25 minutes. Questions follow the six parts of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. Use the atomic masses given in each question. A periodic table and the answer key are at the back.

Part 1 · Solutions and solubility

Q1 SOL-LO1

Match each term to its meaning.

Options: the minor component that is dissolved · the major component · holds the maximum amount of solute · two liquids that mix in any proportion

solute

solvent

saturated

miscible

Q2 SOL-LO1

Will each substance dissolve better in water or in hexane (nonpolar)?

Options: water · hexane

KBr

iodine, I₂

glucose

cooking oil

Q3 SOL-L01

NaCl has a solubility of 36.0 g per 100 mL of water at 25 °C. If 40.0 g is stirred into 100 mL of water at 25 °C, the result is:

- A. an unsaturated solution with no solid left
- B. a saturated solution with 4.0 g of solid left
- C. a supersaturated solution
- D. no solution at all

Part 2 • Percent, ppm and ppb

Q4 SOL-L02

15.0 g of sucrose is dissolved to give 250. g of solution. What is the % m/m?

Q5 SOL-L02

Bleach contains 27.0 g of NaOCl in 500.0 mL of solution. What is its % m/v?

Q6 SOL-L02

What volume of 12.75% m/v glucose solution contains 50.0 g of glucose?

Q7 SOL-L02

An 85 kg body contains 0.012 g of nickel. What is the concentration of Ni in ppm?

Part 3 • Molarity

Q8 SOL-L02

A 350 mL cup of coffee contains 0.150 g of caffeine ($\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$, 194.19 g/mol). What is its molarity? Give the number in units of 10^{-3} M.

Q9 SOL-L02

How many grams of acetic acid ($\text{HC}_2\text{H}_3\text{O}_2$, 60.06 g/mol) are in 0.565 L of a 0.955 M solution?

Q10 SOL-L02

What volume of 0.0882 M formic acid contains 0.888 mol of formic acid?

Q11 SOL-L02

How many mL of 1.84 M H_2SO_4 react with 98.5 g of $\text{Ca}(\text{OH})_2$ (74.10 g/mol)?
 $\text{H}_2\text{SO}_4 + \text{Ca}(\text{OH})_2 \rightarrow 2\text{H}_2\text{O} + \text{CaSO}_4$

Q12 SOL-L02

A solution is 1 mol/L in Ca^{2+} . Its concentration in equivalents is:

- A. 0.5 Eq/L
- B. 1 Eq/L
- C. 2 Eq/L
- D. 4 Eq/L

Part 4 · Dilution

Q13 SOL-L03

A nurse injects 5.00 mL of 0.0500 M heparin into an IV bag, giving a final volume of 255 mL. What is the new concentration? Give the number in units of 10^{-4} M.

Q14 SOL-L03

76.5 mL of 0.885 M KBr is diluted until it is 0.500 M. What is the new volume?

Q15 SOL-L03

In $M_1V_1 = M_2V_2$, the volumes must be in litres.

True / False

Part 5 · Electrolytes and colligative properties

Q16 SOL-L04

Classify each solute.

Options: strong electrolyte · weak electrolyte · nonelectrolyte

KCl

acetic acid

sucrose

NaOH

Q17 SOL-L04

Estimate the freezing point of 0.50 M NaCl (about -1.9 °C per mol of particles per litre). Include the sign.

Q18 SOL-L04

Estimate the boiling point of 1.0 M glucose (about +0.5 °C per mol of particles per litre).

Q19 SOL-L04

Colligative properties depend on:

- A. the identity of the solute
- B. the number of dissolved particles
- C. the colour of the solution
- D. the mass of the solvent only

Part 6 · Osmosis and tonicity

Q20 SOL-L05

What is the osmolarity of 0.20 M Na_2SO_4 ?

Q21 SOL-L05

A solution contains 0.10 M NaCl and 0.10 M glucose. What is its total osmolarity?

Q22 SOL-L05

What happens to a red blood cell placed in each solution? (Blood is about 0.30 osmol.)

Options: no change · swells and bursts (haemolysis) · shrivels (crenation)

0.9% NaCl

pure water

3% NaCl

Pharmacy scenario · Half-normal saline

Scenario

A doctor orders **500 mL of 0.45% m/v NaCl** (half-normal saline). The pharmacy has bags of **0.9% NaCl** (normal saline) and sterile water. (NaCl = 58.44 g/mol)

Q23 SOL-L03

What volume of 0.9% NaCl is needed?

Q24 SOL-L05

What is the osmolarity of 0.45% NaCl? (4.5 g/L)

Q25 SOL-L05

Is 0.45% NaCl isotonic, hypotonic or hypertonic to blood (about 0.30 osmol)?
What will it do to red blood cells, and why is it given slowly and only for special cases?

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

After class: check your understanding

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About 25 minutes. Questions follow the six parts of the lecture and end with a pharmacy scenario.

Answer key

Part 1 · Solutions and solubility

Q1 SOL-L01

Match each term to its meaning.

solute → the minor component that is dissolved; solvent → the major component; saturated → holds the maximum amount of solute; miscible → two liquids that mix in any proportion

Q2 SOL-L01

Will each substance dissolve better in water or in hexane (nonpolar)?

KBr → water; iodine, I₂ → hexane; glucose → water; cooking oil → hexane

Ionic and polar solutes dissolve in water; nonpolar solutes in hexane.

Q3 SOL-L01

NaCl has a solubility of 36.0 g per 100 mL of water at 25 °C. If 40.0 g is stirred into 100 mL of water at 25 °C, the result is:

- A. an unsaturated solution with no solid left
- B. a saturated solution with 4.0 g of solid left**
- C. a supersaturated solution
- D. no solution at all

Only 36.0 g dissolves; the solution is saturated and 4.0 g stays solid.

Part 2 · Percent, ppm and ppb

Q4 SOL-L02

15.0 g of sucrose is dissolved to give 250. g of solution. What is the % m/m?

6 % m/m

$$(15.0 \div 250.) \times 100\% = 6.00\%.$$

Q5 SOL-L02

Bleach contains 27.0 g of NaOCl in 500.0 mL of solution. What is its % m/v?

5.4 % m/v

$$(27.0 \text{ g} \div 500.0 \text{ mL}) \times 100\% = 5.40\% \text{ m/v}.$$

Q6 SOL-L02

What volume of 12.75% m/v glucose solution contains 50.0 g of glucose?

392 mL

$$50.0 \text{ g} \times (100 \text{ mL} \div 12.75 \text{ g}) = 392 \text{ mL}.$$

Q7 SOL-L02

An 85 kg body contains 0.012 g of nickel. What is the concentration of Ni in ppm?

0.14 ppm

$$0.012 \text{ g} \div 85,000 \text{ g} \times 1,000,000 = 0.14 \text{ ppm}.$$

Part 3 · Molarity

Q8 SOL-L02

A 350 mL cup of coffee contains 0.150 g of caffeine ($\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$, 194.19 g/mol). What is its molarity? Give the number in units of 10^{-3} M.

2.21×10^{-3} M

$$0.150 \div 194.19 = 7.72 \times 10^{-4} \text{ mol}; \div 0.350 \text{ L} = 2.21 \times 10^{-3} \text{ M}.$$

Q9 SOL-L02

How many grams of acetic acid ($\text{HC}_2\text{H}_3\text{O}_2$, 60.06 g/mol) are in 0.565 L of a 0.955 M solution?

32.4 g

$\text{mol} = 0.955 \times 0.565 = 0.540 \text{ mol}; \times 60.06 = 32.4 \text{ g}.$

Q10 SOL-L02

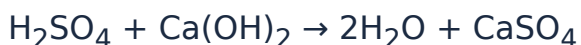
What volume of 0.0882 M formic acid contains 0.888 mol of formic acid?

10.1 L

$\text{L} = \text{mol} \div \text{M} = 0.888 \div 0.0882 = 10.1 \text{ L}.$

Q11 SOL-L02

How many mL of 1.84 M H_2SO_4 react with 98.5 g of $\text{Ca}(\text{OH})_2$ (74.10 g/mol)?



722 mL

$98.5 \div 74.10 = 1.33 \text{ mol}; 1 : 1 \text{ ratio}; 1.33 \div 1.84 = 0.722 \text{ L} = 722 \text{ mL}.$

Q12 SOL-L02

A solution is 1 mol/L in Ca^{2+} . Its concentration in equivalents is:

A. 0.5 Eq/L

B. 1 Eq/L

C. 2 Eq/L

D. 4 Eq/L

Each Ca^{2+} carries 2 charges: $1 \text{ mol/L} \times 2 = 2 \text{ Eq/L}.$

Part 4 • Dilution

Q13 SOL-L03

A nurse injects 5.00 mL of 0.0500 M heparin into an IV bag, giving a final volume of 255 mL. What is the new concentration? Give the number in units of 10^{-4} M.

9.8×10^{-4} M

$$M_2 = (0.0500 \times 5.00) \div 255 = 9.80 \times 10^{-4} \text{ M.}$$

Q14 SOL-L03

76.5 mL of 0.885 M KBr is diluted until it is 0.500 M. What is the new volume?

135 mL

$$V_2 = (0.885 \times 76.5) \div 0.500 = 135 \text{ mL.}$$

Q15 SOL-L03

In $M_1V_1 = M_2V_2$, the volumes must be in litres.

True / False → **False**

Any volume unit works, as long as V_1 and V_2 use the same unit.

Part 5 • Electrolytes and colligative properties

Q16 SOL-L04

Classify each solute.

KCl → strong electrolyte; acetic acid → weak electrolyte; sucrose → nonelectrolyte; NaOH → strong electrolyte

Soluble ionic compounds are strong electrolytes; acetic acid only partly dissociates; sugars stay as molecules.

Q17 SOL-L04

Estimate the freezing point of 0.50 M NaCl (about $-1.9\text{ }^{\circ}\text{C}$ per mol of particles per litre). Include the sign.

$-1.9\text{ }^{\circ}\text{C}$

$0.50 \times 2 = 1.0$ mol particles/L; $1.0 \times 1.9 = 1.9\text{ }^{\circ}\text{C}$ lower: $-1.9\text{ }^{\circ}\text{C}$.

Q18 SOL-L04

Estimate the boiling point of 1.0 M glucose (about $+0.5\text{ }^{\circ}\text{C}$ per mol of particles per litre).

$100.5\text{ }^{\circ}\text{C}$

Glucose gives 1 particle: $1.0 \times 0.5 = 0.5\text{ }^{\circ}\text{C}$ rise: $100.5\text{ }^{\circ}\text{C}$.

Q19 SOL-L04

Colligative properties depend on:

- A. the identity of the solute
- B. the number of dissolved particles**
- C. the colour of the solution
- D. the mass of the solvent only

They depend only on how many particles are dissolved.

Part 6 • Osmosis and tonicity

Q20 SOL-L05

What is the osmolarity of 0.20 M Na_2SO_4 ?

0.6 osmol

$\text{Na}_2\text{SO}_4 \rightarrow 2\text{Na}^+ + \text{SO}_4^{2-}$: 3 particles; $0.20 \times 3 = 0.60\text{ osmol}$.

Q21 SOL-L05

A solution contains 0.10 M NaCl and 0.10 M glucose. What is its total osmolarity?

0.3 osmol

$0.10 \times 2 + 0.10 \times 1 = 0.30$ osmol (add them).

Q22 SOL-L05

What happens to a red blood cell placed in each solution? (Blood is about 0.30 osmol.)

**0.9% NaCl → no change; pure water → swells and bursts (haemolysis);
3% NaCl → shrivels (crenation)**

Pharmacy scenario · Half-normal saline

Q23 SOL-L03

What volume of 0.9% NaCl is needed?

250 mL

$(0.9\%) V_1 = (0.45\%)(500 \text{ mL})$; $V_1 = 250 \text{ mL}$, made up to 500 mL with sterile water.

Q24 SOL-L05

What is the osmolarity of 0.45% NaCl? (4.5 g/L)

0.154 osmol

$4.5 \div 58.44 = 0.0770 \text{ M}$; $\times 2 = 0.154$ osmol.

Q25 SOL-L05

Is 0.45% NaCl isotonic, hypotonic or hypertonic to blood (about 0.30 osmol)?
What will it do to red blood cells, and why is it given slowly and only for special cases?

0.154 osmol is about half of 0.30 osmol, so it is **hypotonic**. Water moves into red blood cells, which swell and could burst (haemolysis) if it is given too fast or too much. It is used carefully, for example to give free water to dehydrated patients with high sodium.

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